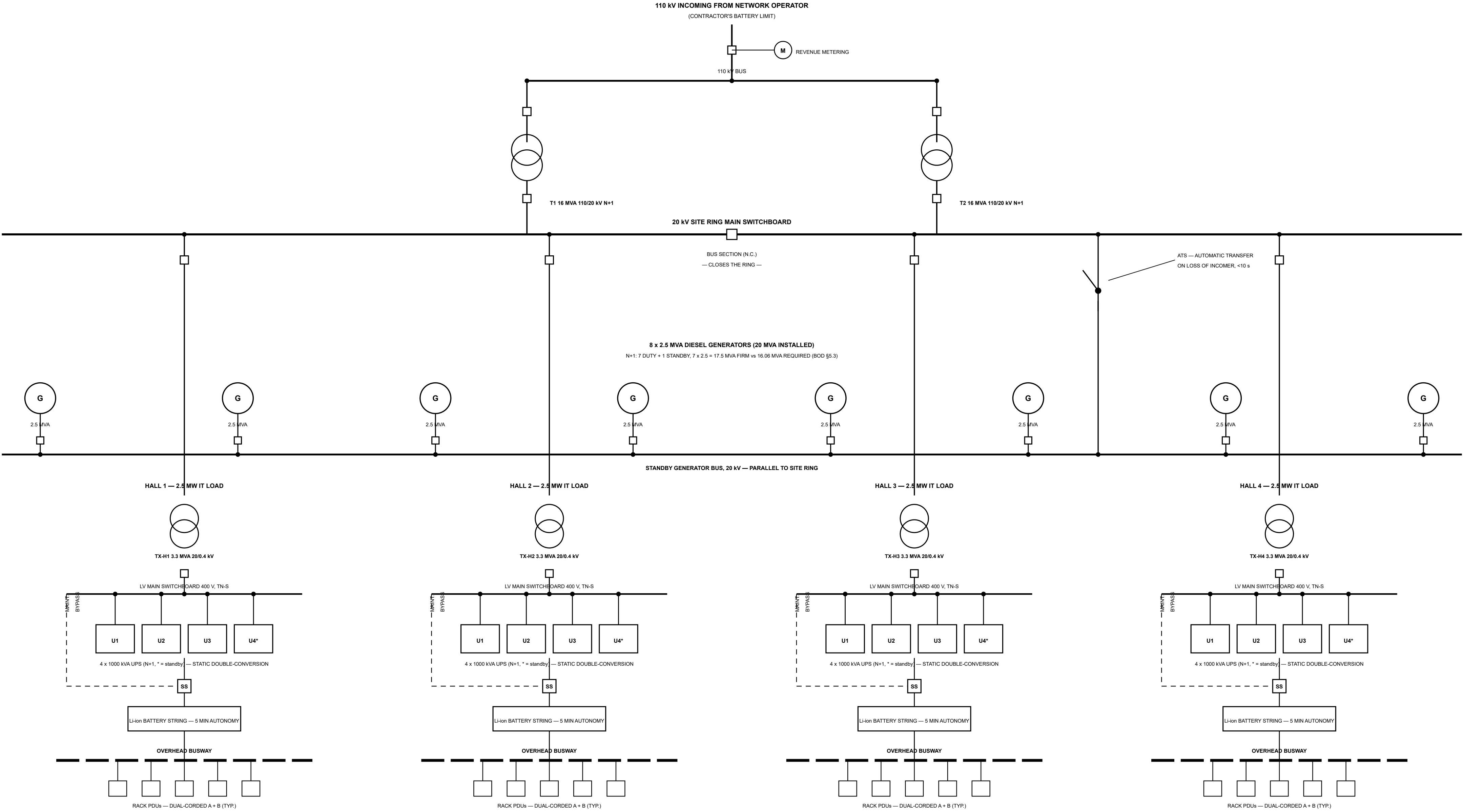




- LEGEND**
- Circuit breaker
 - ⊙ Transformer (two windings)
 - ⊕ Standby diesel generator
 - / Switch / ATS (open shown)
 - Junction — electrical connection
 - (no •) Crossing — NOT connected
 - dashed — Busway
 - SS Static switch (UPS bypass)

- NOTES**
- Redundancy is N+1 at every level from the incoming 110/20 kV transformers to the rack PDU (ER §3.1); any single item of plant is removable for maintenance without loss of IT load.
 - 20 kV switchboard is a ring main with a normally-closed mid-point bus-section breaker, so every hall substation has an alternative supply path via either 110/20 kV transformer.
 - Automatic transfer to standby generation completes within 10 s of loss of incoming supply; UPS batteries carry the load through the transfer (BOD §5.3).
 - UPS: static, double-conversion, N+1 (3 duty + 1 standby) per hall; lithium-ion (LFP) batteries sized for 5 minutes autonomy at full load (BOD §5.2).
 - Earthing: common bonding network, TN-S throughout, single site earth reference bar in the primary substation building.
 - Electrical plant is sized to the PUE guarantee ceiling of 1.25 (25% overhead on IT load) as the conservative sizing basis — see load build-up table, bottom right.

LOAD BUILD-UP — PER HALL (BOD §5.4)		
ITEM	PER HALL (MW)	x4 HALLS (MW)
IT load	2.500	10.000
Cooling (design allowance, 18% of IT)	0.450	1.800
Electrical losses (5% of IT)	0.125	0.500
Lighting & small power (2% of IT)	0.050	0.200
TOTAL PER HALL / SITE SUBTOTAL	3.125	12.500

SITE RECONCILIATION vs 16 MVA PRIMARY INFRASTRUCTURE	
ITEM	VALUE
4 halls (from left table)	12.50 MW
Admin/NOC, external, security (outside PUE boundary)	0.35 MW
TOTAL SITE PEAK (ACTIVE POWER)	12.85 MW
Apparent power @ 0.95 site power factor	13.53 MVA
Firm capacity, dual 16 MVA transformers (N+1, either alone)	16.00 MVA
HEADROOM (growth / diversity / non-coincident peak)	2.47 MVA (~18%)

NOT TO SCALE

PROJECT
POLARIS DC-1 — AVIAPOLIS DATA CAMPUS, PHASE...

EMPLOYER
POLARIS DIGITAL INFRASTRUCTURE OY

CONTRACTOR
NORDBUILD OY

DRAWING
ELECTRICAL SINGLE LINE DIAGRAM

DISCIPLINE
ELECTRICAL

DWG No PDC-DWG-012 SYNTHETIC DEMONSTRATION DATA — NOT A REAL PROJECT	REV A	DATE 24.02.2026	SCALE NTS
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